

ENGINEERING FOUNDATIONS • CLOUD  
TRACK

# Cloud Fundamentals

Foundations of Modern Cloud Computing

~2h 30m / Lecture by: Imran Sayyed

CLOUD STACK

Applications



Platform



Infrastructure

# What we cover today

01

## Foundations

what cloud is, why it exists

02

## Virtualization

VMs, hypervisors, containers

03

## Service Models

IaaS, PaaS, SaaS, FaaS

04

## Deployment

public, private, hybrid

A SHORT HISTORY

# Evolution of computing

1950s

## Mainframes

shared central  
machines



1980s

## Personal Computers

one machine per  
desk



2000s

## Data Centers

racks of servers



**2010s+ Cloud Computing** on-demand resources over the internet

THE OLD WAY HURTS

# Problems with traditional infrastructure

## High Cost

buy servers up front  
before any users  
arrive

## Maintenance

staff to patch, cool,  
and repair hardware

## Scaling

weeks to add  
capacity for a traffic  
spike

## Downtime

one failed machine  
can stop the  
business

## CORE DEFINITION

# What is cloud computing?

Cloud computing is the delivery of computing resources **servers, storage, networking, and software** over the internet, on demand, paying only for what you use.

on demand

over the internet

pay per use



YOU ALREADY USE IT

# Cloud computing, explained simply

## Netflix

Movies stream from servers you never see or own.

## Google Drive

Files live online, reachable from any device.

## Gmail

Email runs in a browser, nothing installed locally.

# Characteristics of cloud computing

## On Demand

request resources instantly

## Self Service

no human in the loop

## Elasticity

grow and shrink automatically

## Scalability

handle huge load when needed

## Availability

running around the clock

## Pay As You Go

billed only for usage

BUILDING BLOCKS

# What makes up a computer?

layer 01

## Hardware

CPU, RAM, disk, the physical machine

layer 02

## Operating System

manages and shares the hardware

layer 03

## Networking

connects the machine to others

layer 04

## Storage

keeps data and files persistent



THE BIG SHIFT

# Why businesses moved to cloud

## Traditional

Buy hardware up front

Weeks to add capacity

You run the data center

Pay even when idle

## Cloud

Rent what you need

Scale in minutes

Provider runs the hardware

Pay only for usage

WHY IT WINS

# Cloud benefits

## Cost

no upfront  
hardware, pay per  
use

## Speed

launch servers in  
minutes

## Scalability

grow with demand  
instantly

## Global Reach

deploy near users  
worldwide

WHERE CLOUD LIVES

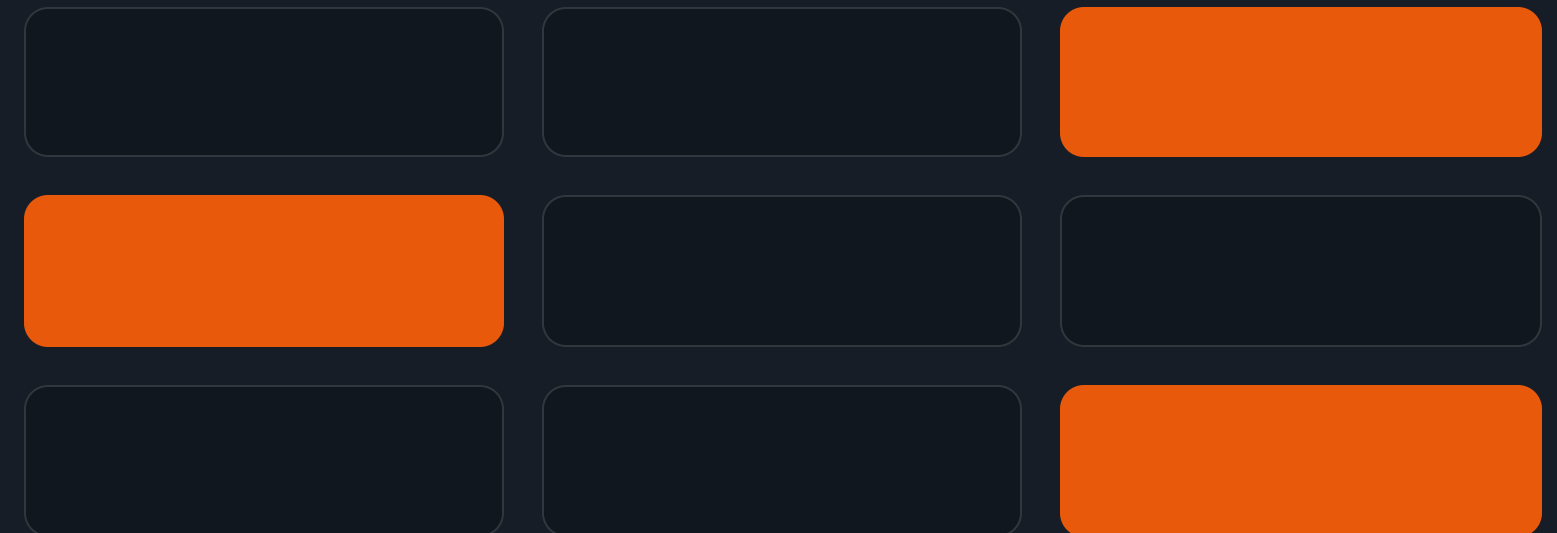
# Data center introduction

A data center is a building full of servers, storage, cooling, and power that runs the cloud behind the scenes.

power

cooling

network



THE RAW MACHINE

# Physical servers

## Dedicated

one company owns the whole box

## Powerful

strong CPU, large memory and disk

## Fixed

capacity is set by the hardware bought

WASTED CAPACITY

# Why physical servers are inefficient

TYPICAL USAGE

15%

of one server's power is actually used  
most of the time

**One app per server**

most of the box sits idle

**Always powered**

full power and cooling cost even when idle

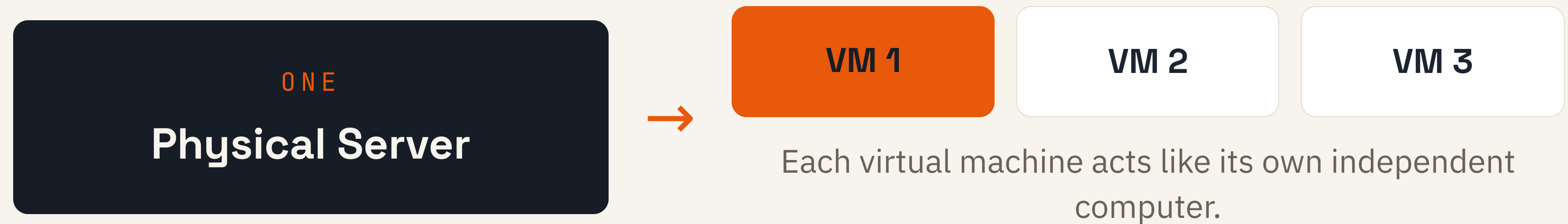
**Hard to share**

spare capacity cannot be reused easily



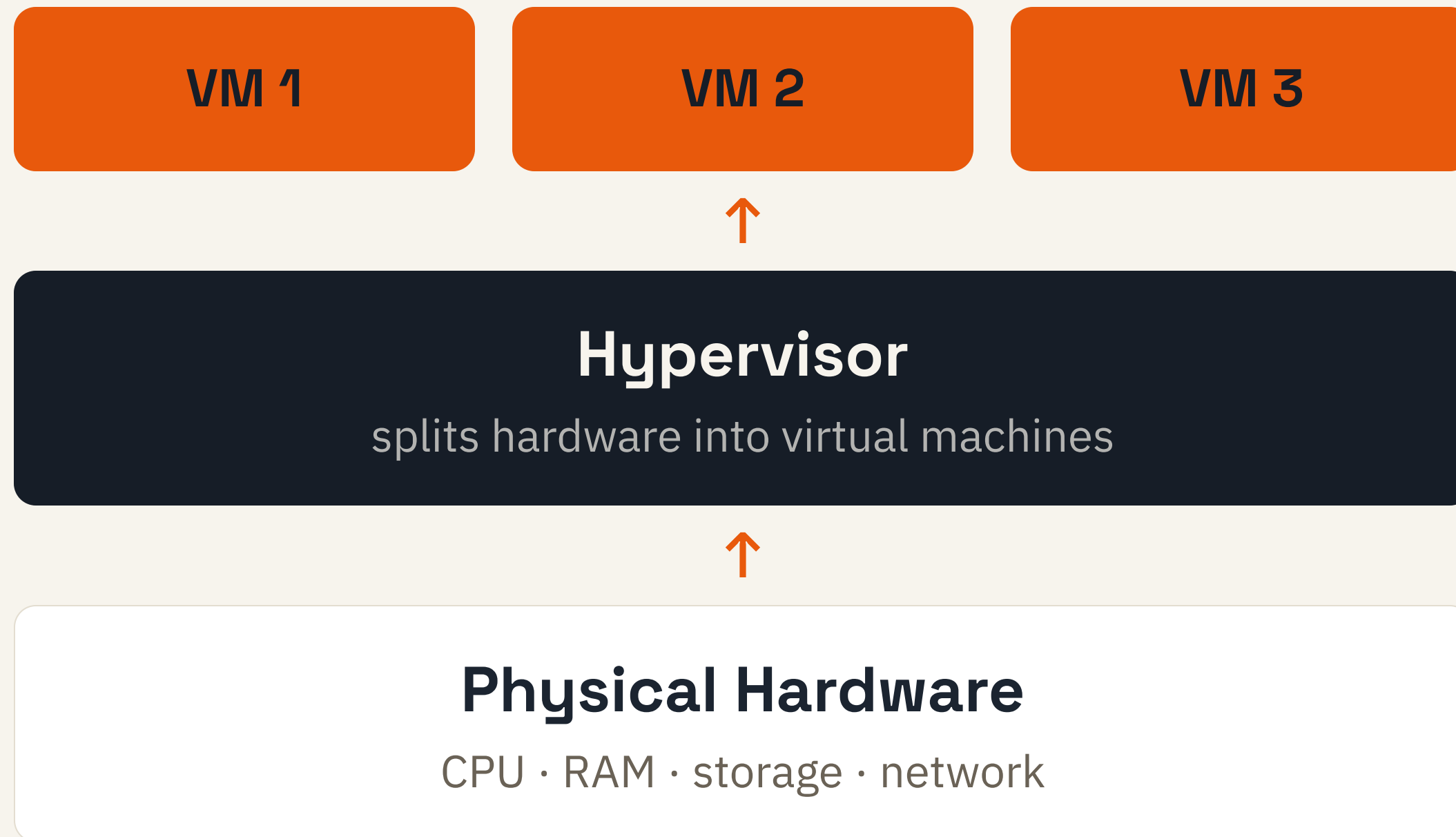
ONE BOX, MANY MACHINES

# Introduction to virtualization



THE STACK

# Virtualization architecture



WHY IT MATTERS

# Benefits of virtualization

## Better utilization

one machine runs many workloads

## Isolation

a crash in one VM never touches the others

## Fast provisioning

spin up a new server in minutes

## Lower cost

fewer physical machines to buy and power



THE MANAGER

# What is a hypervisor?

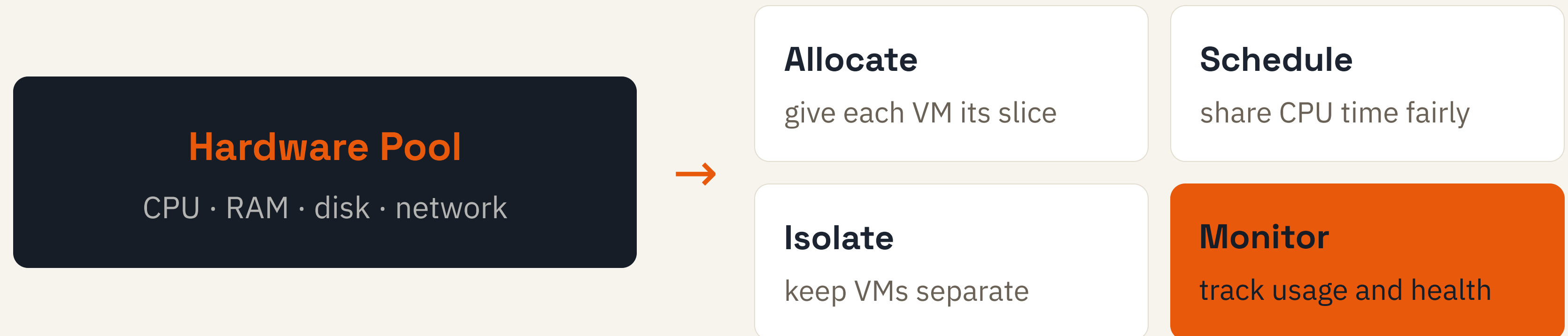
A hypervisor is the software layer that creates and runs virtual machines, sharing one set of hardware among many of them.

creates VMs

shares hardware

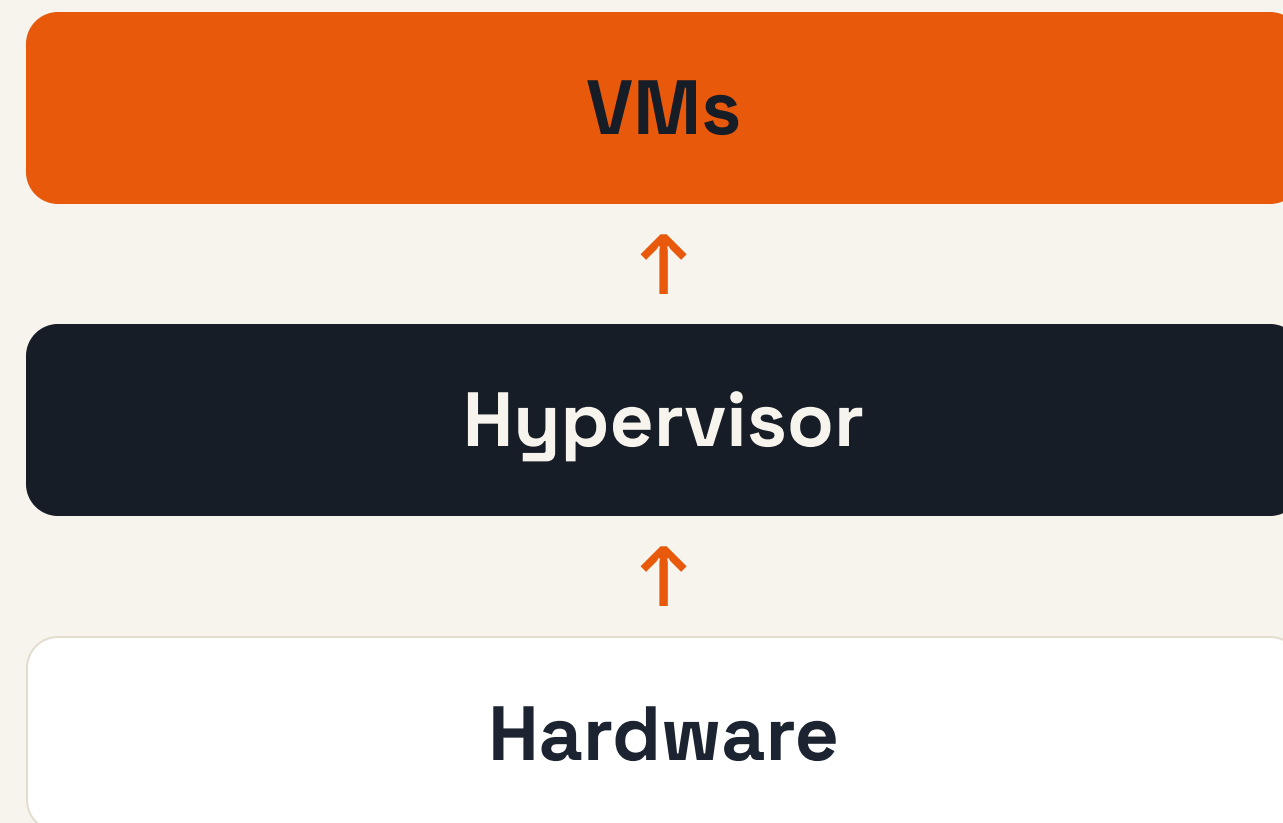
keeps them isolated

# Hypervisor responsibilities



BARE METAL

# Type 1 hypervisor



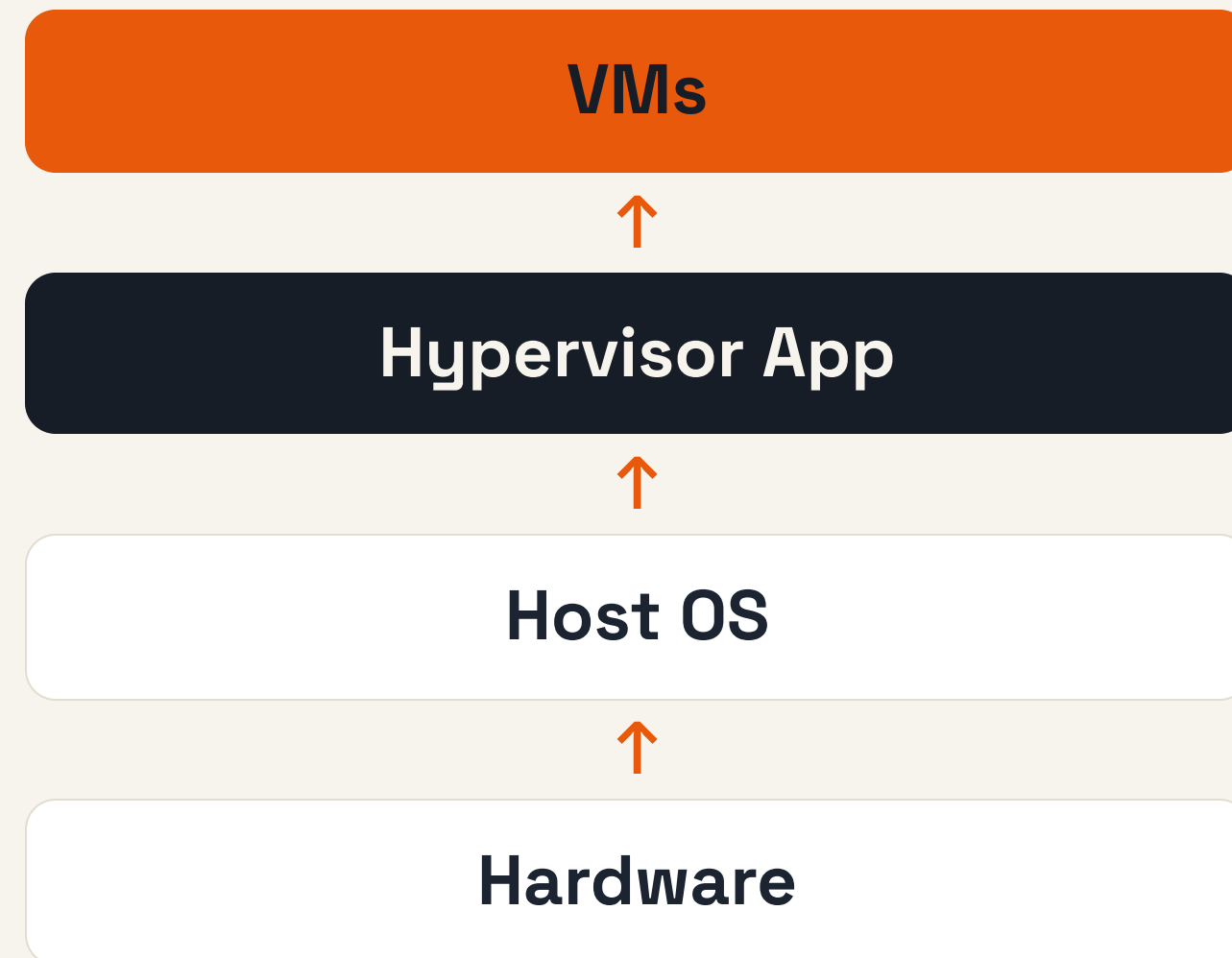
Runs directly on the hardware, with no host operating system in between.

EXAMPLES

ESXi · Hyper-V · KVM

HOSTED

# Type 2 hypervisor



Runs as an app on top of a normal operating system, like any other program.

## EXAMPLES

VirtualBox • VMware Workstation

SIDE BY SIDE

# Hypervisor comparison

| Aspect      | Type 1        | Type 2           |
|-------------|---------------|------------------|
| Runs on     | bare hardware | a host OS        |
| Performance | high          | moderate         |
| Best for    | data centers  | laptops, testing |



V I R T U A L   O R   R E A L

# Virtual machines vs physical machines

## Physical Machine

One operating system

Fixed, dedicated hardware

Slow to provision

Hard to move or copy

## Virtual Machine

Many on one machine

Shares a hardware pool

Created in minutes

Easy to clone and migrate

LIGHTER THAN VMS

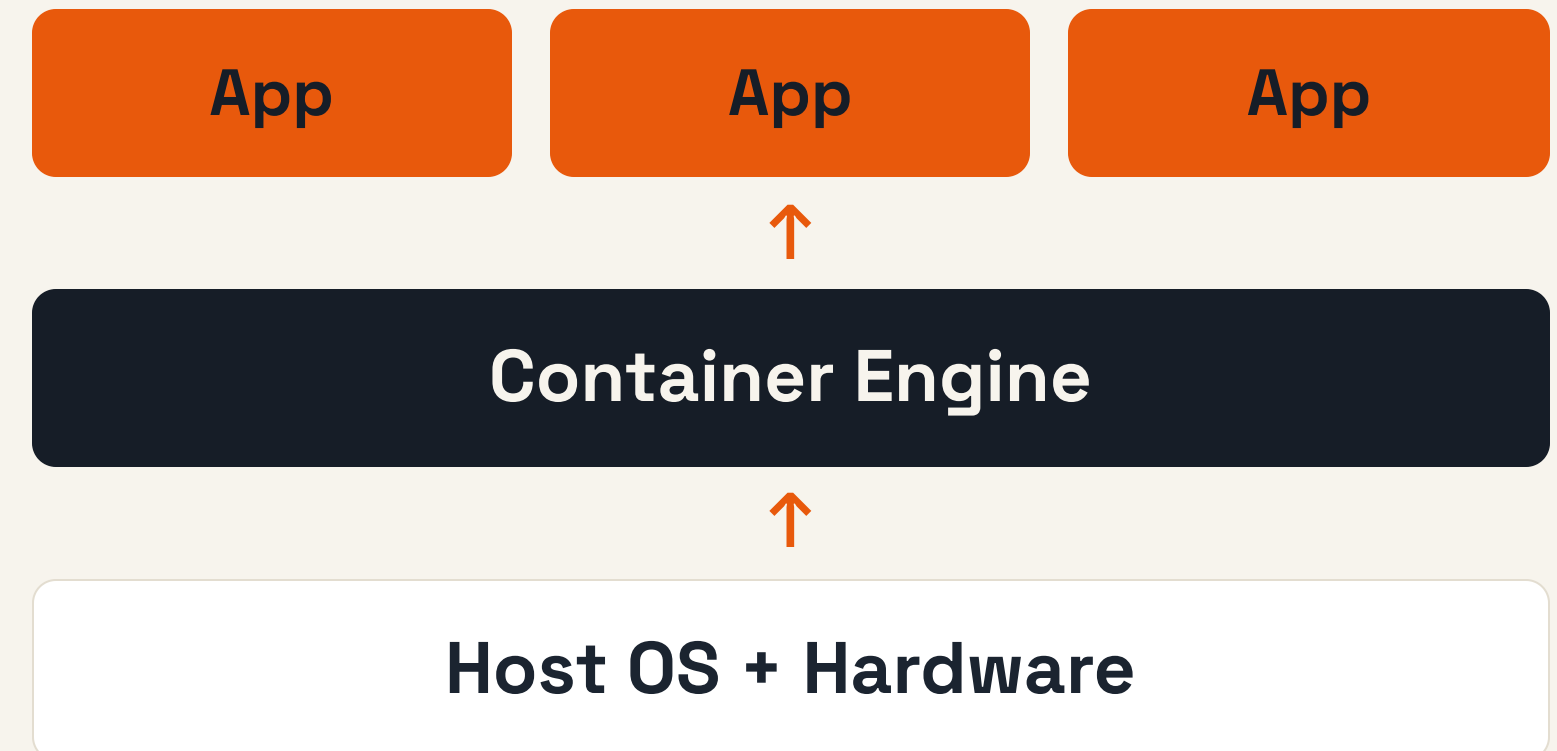
# Containers introduction

A container packages an app with everything it needs and shares the host operating system instead of running a full one.

lightweight

fast start

portable



## KEY COMPARISON

# VM vs container

## Virtual Machine

App

Guest OS

**Hypervisor**

Hardware

Each VM ships a full guest OS, so it is heavier and slower to start.

## Container

App

**Container Engine**

Shared Host OS

Hardware

Containers share the host OS, so they are small and start in seconds.



NO SERVERS TO MANAGE

# Introduction to serverless

With serverless, you write code and the provider runs it on demand. Servers still exist, but you never see or manage them.

## No setup

no servers to provision

## Auto scale

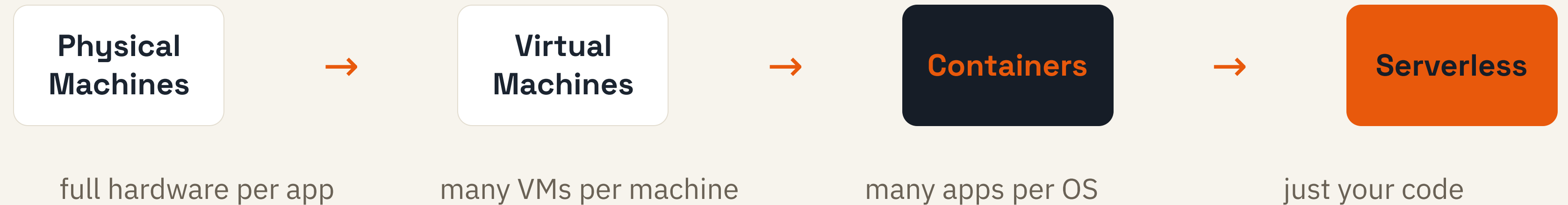
handles one or millions  
of calls

## Pay per run

billed only while code  
runs

FROM HEAVY TO LIGHT

# Evolution of compute



HOW MUCH YOU MANAGE

# Service models overview

**IaaS**

rent the  
infrastructure

**PaaS**

rent the platform

**SaaS**

use the software

**FaaS**

run just functions

you manage more



provider manages more

WHO DOES WHAT

# Shared responsibility

## Provider manages

Data centers and hardware

Networking and storage

Virtualization layer

## You manage

Your apps and data

Access and permissions

Configuration choices

The higher the service model, the more the provider handles for you.

INFRASTRUCTURE AS A SERVICE

# laaS



You rent raw compute, storage, and networking, then build everything on top yourself.

## EXAMPLES

AWS EC2 · Compute Engine · Azure VM



IN PRACTICE

# laaS example

A startup needs servers for a new app but does not want to buy hardware.

## Rent a VM

spin up EC2 in minutes



## Install stack

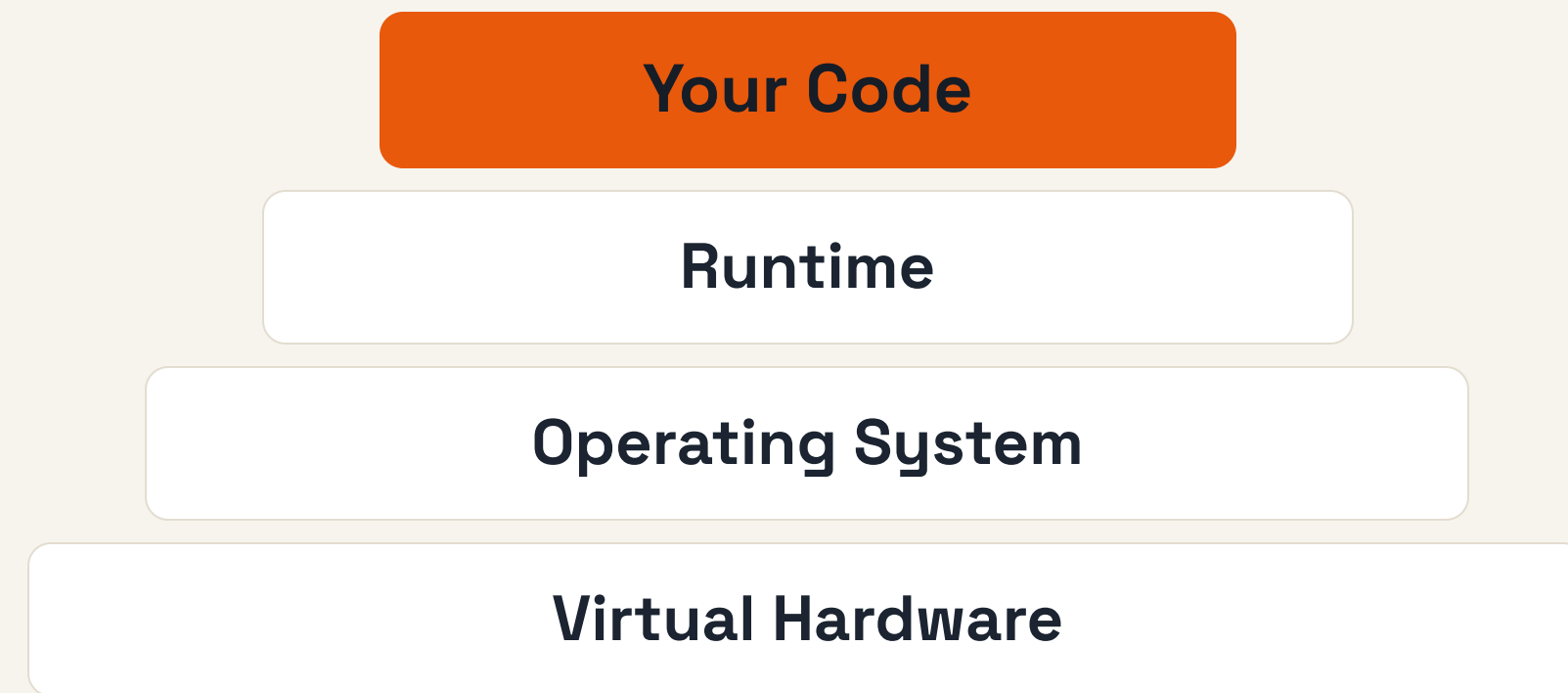
OS, runtime, database

## Scale on demand

add servers as traffic  
grows

PLATFORM AS A SERVICE

# PaaS



You deploy your code and the provider runs the platform underneath it for you.

## EXAMPLES

App Engine · Azure App Service · Heroku

IN PRACTICE

# PaaS example

A developer wants to ship a web app without managing servers or operating systems.

**Write code**

build the app locally



**Push to deploy**

platform builds and runs  
it

**Auto managed**

scaling and patching  
handled



SOFTWARE AS A SERVICE

# SaaS

**Ready-to-Use App**



provider runs everything below

You just log in and use the software. The provider handles all the layers beneath it.

## EXAMPLES

Gmail · Office 365 · Zoom

IN PRACTICE

# SaaS example

A team needs email and video calls without installing or maintaining any software.

**Open a browser**

no install needed



**Sign in**

use it instantly

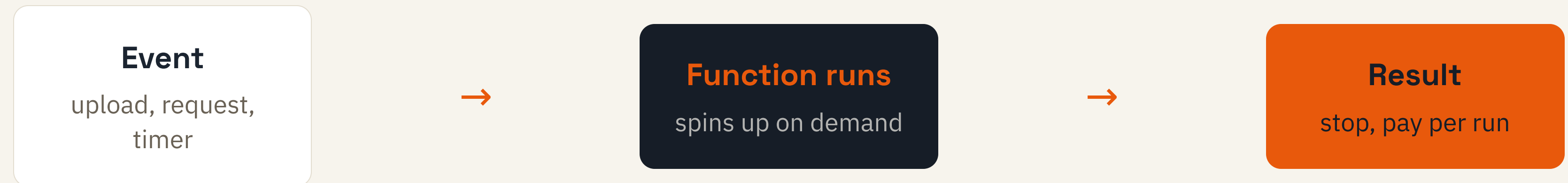
**Always updated**

provider patches it for  
you

FUNCTIONS AS A SERVICE

# FaaS

You upload small functions that run only when an event triggers them.



examples: AWS Lambda · Cloud Functions · Azure Functions

ALL TOGETHER

# Service model comparison

| Model | You manage           | Example    |
|-------|----------------------|------------|
| IaaS  | OS, runtime, apps    | AWS EC2    |
| PaaS  | just your code       | App Engine |
| SaaS  | nothing, just use it | Gmail      |
| FaaS  | single functions     | AWS Lambda |

WHERE THE CLOUD LIVES

# Deployment models

## Public

shared infrastructure run by a provider

## Private

dedicated to a single organization

## Hybrid

a mix of public and private



SHARED BY MANY

# Public cloud

Infrastructure owned by a provider and shared across many customers over the internet.

No hardware to buy

Pay only for what you use

Scales almost instantly

AWS • Azure • Google Cloud

## Shared Provider Cloud

Company A

Company B

Startup C

User D

FOR ONE ORGANIZATION

# Private cloud

## Single Organization

Team A

Team B

Team C

Infrastructure dedicated to one organization, for more control and security.

Full control over resources

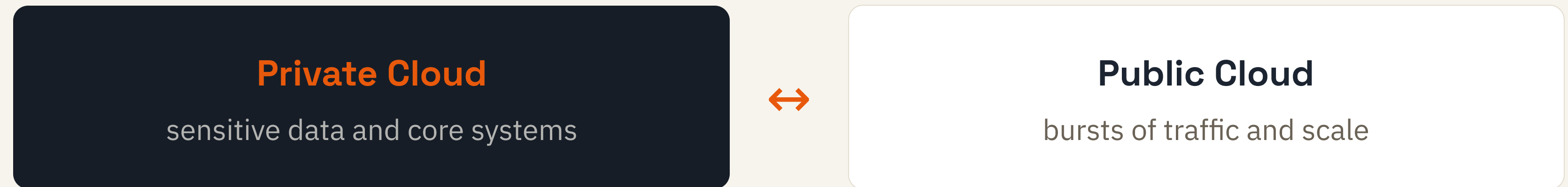
Strong privacy and compliance

Higher cost to run

BEST OF BOTH

# Hybrid cloud

A mix of private and public cloud, connected so workloads can move between them.



Keep critical data private while using the public cloud for extra capacity.



SIDE BY SIDE

# Deployment comparison

| Aspect   | Public             | Private          | Hybrid            |
|----------|--------------------|------------------|-------------------|
| Cost     | low                | high             | balanced          |
| Control  | low                | high             | flexible          |
| Best for | startups, web apps | banks, hospitals | large enterprises |

YOU USE IT DAILY

# Cloud in real life

## Netflix

streams video from the public cloud at global scale

## Google Photos

stores and backs up your photos in the cloud

## Spotify

delivers music on demand from cloud servers

## Online Banking

runs securely on private and hybrid clouds

TIE IT TOGETHER

# Revision mind map

**Cloud Computing**

```
graph TD; CC[Cloud Computing] --> V[Virtualization]; CC --> SM[Service Models]; CC --> D[Deployment]; CC --> B[Benefits];
```

**Virtualization**

VMs, hypervisors,  
containers

**Service Models**

IaaS, PaaS, SaaS, FaaS

**Deployment**

public, private, hybrid

**Benefits**

scale, cost, speed

CHECK YOURSELF

# Quick quiz

Q1

What does a hypervisor do?

Q2

How is a container different from a VM?

Q3

Which model is Gmail: IaaS, PaaS, or SaaS?

Q4

When would you choose a hybrid cloud?

WRAP UP

# Key takeaways

## **Cloud is on-demand computing**

resources over the internet, billed by use

## **Virtualization makes it possible**

VMs and containers share hardware efficiently

## **Service models set the line**

IaaS to SaaS shift work to the provider

## **Deployment fits the need**

public, private, or hybrid by priority